

Your A/B test “won.” Was it real, or just luck?

A landing page test for an online Excel course:
28,905 visitors, 28 days, two versions of the page.

Frequentist

Bayesian

Permutation

Segments

Swipe to see what the data really said →

THE HEADLINE

The new page converted more visitors

Version A · current page

2.66%

Version B · new page

3.11%

+17% relative lift in sign-ups

Most reports stop here.

But is a gap this size bigger than random chance?

IS IT REAL?

I didn't trust one p-value. I checked five ways.

Split health check

Was traffic really 50/50?

✓ 50.2 / 49.8

Two-proportion z-test

The classic test

$p = 0.022$

Permutation test

10,000 random label shuffles

$p = 0.024$

Bayesian model

Probability B beats A

98.9%

Regression with controls

Device, channel, weekday

+17.5% odds

Every method agrees: the lift is very unlikely to be luck.

WHO DOES IT HELP?

Almost the entire win came from mobile



Mobile

+37%



2.09% → 2.87% · still significant after correcting for multiple tests



Desktop

-2%



3.62% → 3.54% · no real change

Tablet

unclear



Too few visitors to judge either way

Makes sense: B's shorter sign-up form is much easier to fill in on a phone.

TRAP #1

Checking results every day creates fake winners

I simulated 2,000 tests where both pages were identical, so any "winner" is fake.

Checks once, at the
planned end date

4%

false winners

Peeks daily, stops at
the first "win"

23%

false winners

Same data, same 5% threshold. Impatience alone makes false wins **~5× more likely**. Fix the sample size and end date before you start.

TRAP #2

More sign-ups doesn't mean more money

Buyers chose between a \$49 Standard course and a \$129 Pro bundle.

A buyers

Pro 35%

Standard 65%

B buyers

Pro 26%

Standard 74%

Revenue per visitor

\$2.04 → \$2.18

The 95% range runs from **-\$0.17 to +\$0.45** per visitor. B wins more students, but whether it earns more money is **not yet proven**.

THE DECISION

Not just "ship B." A plan the business can act on.

1

Ship B on mobile now

Strong, proven lift with negligible risk

2

Keep testing on desktop

No evidence B helps there yet

3

Fix the upsell first

Then re-test with revenue as the main metric

Full report, code and data: link in the post

github.com/Manjirigajmal/ab-test-analysis-r

Built on a simulated dataset so the full method can be shared.